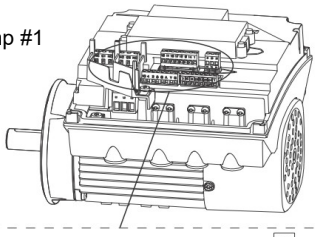
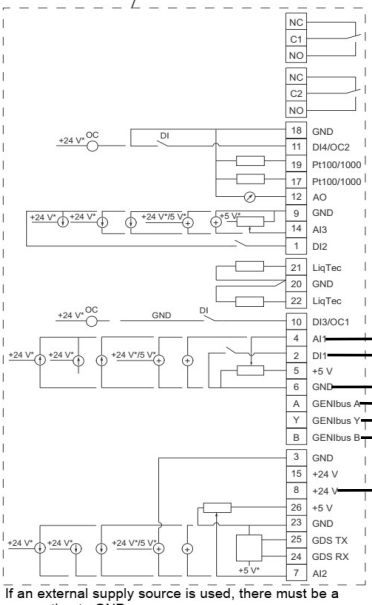
Grundfos/PACO Permanent Magnet MLE motor (model H,I or J) (up to 15 HP)
Multi Pump Operation:
-Each MLE must be connected to a pressure sensor or differential pressure sensor
Each connected pump & MLE motor must be identical model and size
-All low voltage wiring requires 16ga twisted shielded pair
-All field wiring to be done by the contractor

Pump #1



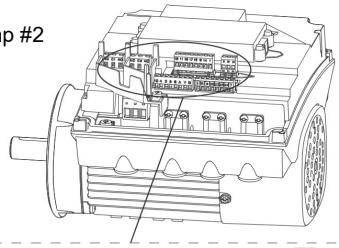


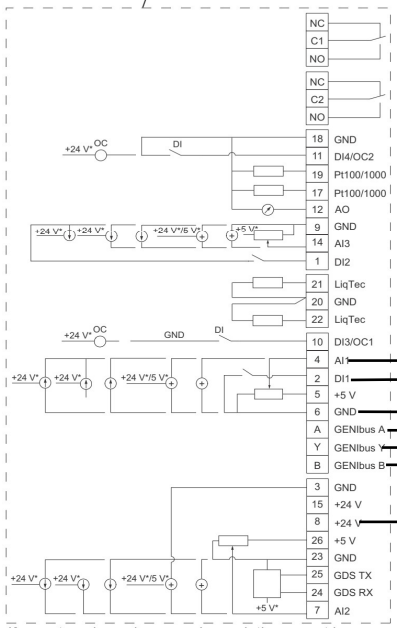
* If an external supply source is used, there must be a connection to GND.

Pressure sensor (Analog input):
0-20 mA / 4-20 mA
0.5 - 3.5 V / 0-5 V / 0-10 V

Bus Connection

Pump #2





* If an external supply source is used, there must be a connection to GND.

Start/Stop Relay (by others) - Normally open - 2-pole - Closed when AC units are in operaton. AC units to be wired in parallel.

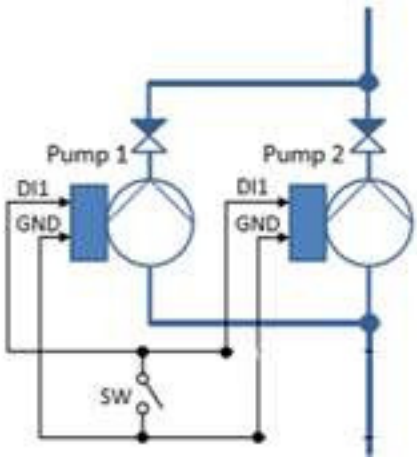
Pressure sensor (Analog input):
0-20 mA / 4-20 mA
0.5 - 3.5 V / 0-5 V / 0-10 V

Electrical Start/Stop Connections for Multi Pump

Remove the jumper (terminals 2 & 6), because we will use the digital input 1 as an external start-stop.

Terminal	Function
2	Digital Input 1
6	Ground

The multi pump system is to be controlled by an external, contractor provided start/stop switch as shown below. The switch can also be the a single or multiple wired in parallel normally open start/stop relay outputs from the system AC units.



The Digital Input 1 (terminal #2) terminals are to be wired together.
The Ground (terminal #6) terminals are to be wired together. (Must be #6 on both pumps)
The Switch (SW) is to be placed across these two wires.
The external switch (SW) will start the complete system, no matter which pump has the master (duty) role.
The digital input 1 on both of the pumps in the system is to be set to "Ext stop".
The factory setting of digital input 1 is "Ext stop".
The switch (SW) is to be closed for the pumps to run.

Connect the AYB terminals of pump number 1 to the AYB terminals of pump number 2.

Use a screened cable (16 ga twisted shielded pair) between the GENIbus terminals. Connect the screen of the cable to the frame.



Multi Pump Function

The multi pump function enables the control of two pumps connected in parallel without the use of external controllers.

The pumps in a multi pump system communicate with each other via the wireless GENIair connection or via the wired GENI connection.

Three different multi pump functions are available:

- Alternating operation, time – NOT RECOMMENDED
- Alternating operation, energy – RECOMMENDED SET-UP OPTION
- Back-up operation – NOT RECOMMENDED

Alternating operation functions as a duty/standby operating mode with only one pump in operation at a time.

Alternating operation is possible with two pumps of the same size, type, and connected in parallel.

Any pump can function as the master (duty) pump and take over the master (duty) pump function if the master (duty) pump should fail. This is only possible if each pump is configured with a dedicated pressure sensor.

Installation - Confirm

Each pump requires a non-return (check) valve in series with the pump.

Ensure the pumps are correctly installed in the system with non-return (check) valves.

Enable the Multi Pump Function

The main purpose of the multi pump function is to ensure an even amount of running hours and to ensure that the standby pump takes over if the running pump stops due to an alarm.

The operating pump can be alternated based on load, alternation takes place when the duty pump has consumed 20kWh.

Prepare the second pump.

Electrical Connections
Run the start-up guide / Assisted pump setup
Set Units; Display brightness / Blank display
Prime the pump

Give the second pump the pump number “2” using the graphical control panel

Home menu
Settings menu
Communication OK
Pump number OK
OK (change to Pump 2) OK
Home menu

Give the first pump the pump number “1” using the graphical control panel

Home menu
Settings menu
Communication OK
Pump number OK
OK (change to Pump 1) OK
Home menu

Go to the “setup of the multi-pump system” using pump number 1.

Home menu
Assist
Setup of the multi-pump system OK
>
Wired GENIbus OK > (Select the communication channel)
>
Alternating operation, energy OK > (Select the multi-pump function)

OK	(Search for other pumps)
OK >	(Pump 2 winks)
>	(See * below)
OK	(Confirm the setting)

* The pump changeover will be made automatically when the difference in energy consumption of the two pump is equal to the daily energy consumption of one pump running at the rated performance

Below is the graphical control panel displays for the two pumps.

Pump 1



Pump 2



The symbol at the end of the white arrow indicates master (duty) pump.
The symbol at the end of the yellow arrow indicates slave (standby) pump.
The symbol at the end of the red arrows indicates the pumps are turned off by the start stop button.

Operate the pumps in the multi-pump mode by pressing the start stop button on pump one and then press the start stop button on pump 2.

Pump 1 operates, and pump 2 is in standby mode.

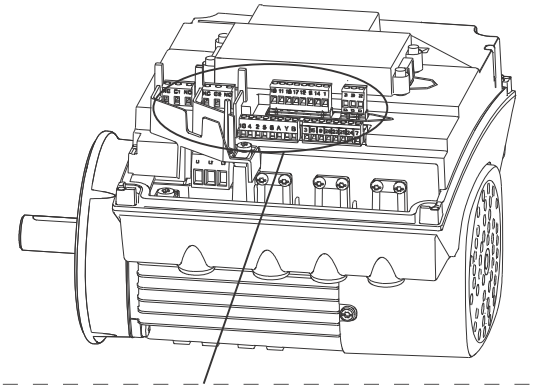
The graphical displays of pump 1 and pump 2 look as follows:

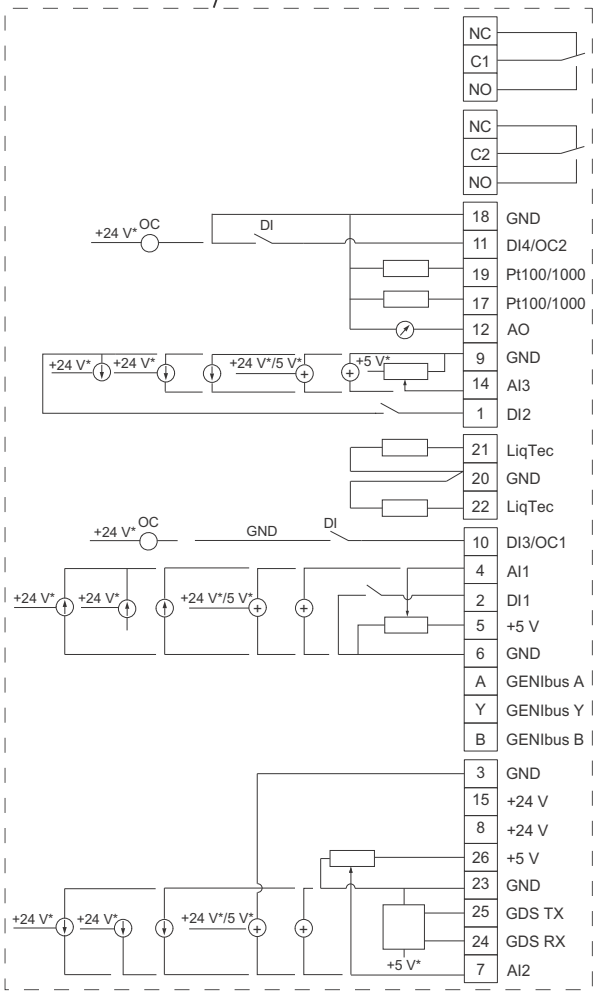
Pump 1



Pump 2







* If you use an external supply source, there must be a connection to GND.

Fig. 8 Connection terminals, CRE, CRIE, CRNE, CRKE, SPKE, MTRE and CME pumps

Terminal	Type	Function
NC	Normally closed contact	Signal relay 1 (LIVE or PELV)
C1	Common	
NO	Normally open contact	
NC	Normally closed contact	Signal relay 2 (PELV only)
C2	Common	
NO	Normally open contact	
18	GND	Ground
11	DI4/OC2	Digital input/output, configurable. Open collector: Max. 24 V resistive or inductive.
19	Pt100/1000 input 2	Pt100/1000 sensor input
17	Pt100/1000 input 1	Pt100/1000 sensor input
12	AO	Analog output: 0-20 mA / 4-20 mA 0-10 V
9	GND	Ground
14	AI3	Analog input: 0-20 mA / 4-20 mA 0-10 V
1	DI2	Digital input, configurable
21	LiqTec sensor input 1	LiqTec sensor input (white conductor)
20	GND	Ground (brown and black conductors)
22	LiqTec sensor input 2	LiqTec sensor input (blue conductor)
10	DI3/OC1	Digital input/output, configurable. Open collector: Max. 24 V resistive or inductive.
4	AI1	Analog input: 0-20 mA / 4-20 mA 0.5 - 3.5 V / 0-5 V / 0-10 V
2	DI1	Digital input, configurable
5	+5 V	Supply to potentiometer and sensor
6	GND	Ground
A	GENIbus, A	GENIbus, A (+)
Y	GENIbus, Y	GENIbus, GND
B	GENIbus, B	GENIbus, B (-)
3	GND	Ground
15	+24 V	Supply
8	+24 V	Supply
26	+5 V	Supply to potentiometer and sensor
23	GND	Ground
25	GDS TX	Grundfos Digital Sensor output
24	GDS RX	Grundfos Digital Sensor input
7	AI2	Analog input: 0-20 mA / 4-20 mA 0.5 - 3.5 V / 0-5 V / 0-10 V

7.44 "Multi-pump setup" ("Setup of multi-pump system")

Pump variant	"Multi-pump setup" ("Setup of multi-pump system")
CME	•
CRE, CRIE, CRNE, CRKE, SPKE, MTRE	•

The multipump function enables the control of two pumps connected in parallel without the use of external controllers. The pumps in a multipump system communicate with each other via the wireless GENIair connection or the wired GENI connection.

You can set a multipump system via the master pump, that is the first selected pump.

If two pumps in the system are configured with an outlet-pressure sensor, they can all function as master pumps and take over the master pump function if the other should fail. This provides additional redundancy in the multipump system.

The multipump functions are described in the following sections.

7.44.1 "Alternating operation"

Alternating operation functions as a duty/standby operating mode and is possible with two pumps of same size and type connected in parallel. The main purpose of the function is to ensure an even amount of running hours and to ensure that the standby pump starts if the duty pump stops due to an alarm.

Each pump requires a non-return valve in series with the pump.

You can choose between two alternating operation modes:

"ALTERNATING OPERATION, ENERGY" IS RECOMMENDED

- "Alternating operation, time"
  Change from one pump to the other is based on time.
- "Alternating operation, energy"
  Change from one pump to the other is based on energy consumption.

If the duty pump fails, the other pump starts.

7.44.2 "Backup operation"

Backup operation is possible with two pumps of same size and type connected in parallel. Each pump requires a non-return valve in series with the pump.

One pump is operating continuously. The backup pump is operated for a short time each day to prevent seizing up. If the duty pump stops due to a fault, the backup pump starts.

7.44.3 Setting up a multipump system

You can set a multipump system in the following ways:

- *Grundfos GO and wireless pump connection*
- *Grundfos GO and wired pump connection*
- *Advanced control panel and wireless pump connection*
- *Advanced control panel and wired pump connection*

See step-by-step descriptions below.

Grundfos GO and wireless pump connection

1. Power on both pumps.
2. Establish contact to one of the pumps with Grundfos GO.
3. Set the needed analog and digital inputs via Grundfos GO according to the connected equipment and the required functionality. See section *7.41 "Assisted pump setup"*.
4. Assign a pump name to the pump using Grundfos GO. See section *7.35 "Pump name"*.
5. Disconnect Grundfos GO from the pump.
6. Establish contact to the other pump.
7. Set the needed analog and digital inputs via Grundfos GO according to the connected equipment and the required functionality. See section *7.41 "Assisted pump setup"*.
8. Assign a pump name to the pump using Grundfos GO. See section *7.35 "Pump name"*.
9. Select the "Assist" menu and choose "Setup of multi-pump system".
10. Select the desired multipump function. See section *7.44.1 "Alternating operation"* and *7.44.2 "Backup operation"* above.
11. Press [>] to continue.
12. Set the time for pump changeover, that is the time at which the alternation between the two pumps is to take place.



This step applies only if you have selected "Alternating operation, time" and if the motors are fitted with FM 300.

13. Press [>] to continue.
14. Select "Radio" as the communication method to be used between the two pumps.
15. Press [>] to continue.
16. Press "Select pump 2".
17. Select the pump from the list.



Use the [OK] or button to identify the pump.

18. Press [>] to continue.
19. Confirm the setting by pressing [Send].
20. Press [Finish] in the "Setup complete" dialog box.
21. Wait for the green indicator light in the middle of Grundfos Eye to light up.